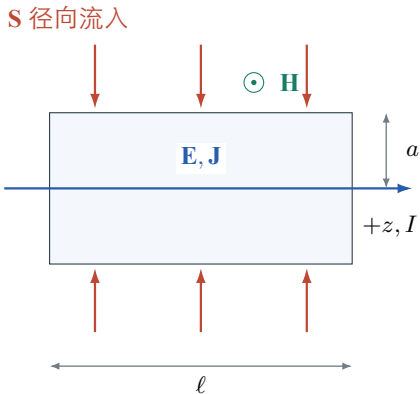
M7-006 圆柱电阻表面的能流



$$E_z = I/(\sigma\pi a^2)$$

$$\mathbf{S}(a) = -\frac{I^2}{2\sigma\pi^2 a^3}\mathbf{e}_r$$

对侧表面 $2\pi a\ell$ 积分，得到 $P_{\text{in}} = I^2R$。